DAILY EDITORIAL SUMMARY

24th April 2026

Topics to be covered:

- ✓ **High Heat: On India and Warming**
- ✓ **Scaling Climate Adaptation From Policy to Grassroots**

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Topics

High Heat: On India and Warming

GS III: Environmental Pollution and degradation

Context:

India is currently facing a severe heat crisis, with temperatures exceeding **40°C in at least eight states**. This has led to a significant loss of livelihoods, particularly for outdoor workers, resulting in an estimated **247 billion working hours lost in 2024**.

What is a Heatwave? (IMD Definitions)

The India Meteorological Department (IMD) defines heatwaves based on geographical regions:

- **Plains:** When the maximum temperature reaches 40°C and is **4.5°C to 6.4°C above normal**. If it exceeds 6.4°C above normal (totaling over 46.5°C), it is a **severe heatwave**.
- **Coastal Regions:** When temperatures exceed **37°C**.
- **Hilly Areas:** When temperatures exceed **30°C**.

Conditions and Causes

Five primary factors are driving the current extreme heat:

- **El Niño Effect:** Warming in the Pacific Ocean weakens the Indian monsoon, leading to higher temperatures.
- **Weak Western Disturbances:** The absence of winter and spring rainfall (Maowat) has removed a natural cooling effect.
- **Weak Convective Currents:** A lack of local air rising and condensing into clouds has prevented natural afternoon cooling.
- **Urban Heat Islands:** Concrete and asphalt in cities trap heat during the day and fail to release it at night, keeping urban areas hotter than surroundings.
- **Wet Bulb Temperature:** A deadly combination of **high heat and high humidity** that prevents sweat from evaporating, making even 35°C lethal.

About Heat Action Plans (HAPs):

- Currently, Heat Action Plans in India are primarily **localized and implemented at the state or district level** rather than through a standardized national approach.

Challenges:

- **Reactive Approach:** Efforts usually begin only after a heat event occurs rather than being proactive.
- **Funding Shortages:** There is a lack of long-term, permanent funding for heat mitigation.
- **Informal Sector Vulnerability:** Workers in the informal sector face the brunt of health risks and income loss with little protection.
- **Lack of Urban Regreening:** Cities continue to expand without sufficient green spaces to counter the heat island effect.

Way Forward:

- **Proactive Planning:** Implementing measures before the crisis hits.
- **Permanent Infrastructure:** Establishing **mobile health units** and heat safety legislation to protect workers.
- **Urban Greening:** Actively planting trees and restoring local ecology in cities.
- **International Collaboration:** Joining groups like the **Columbia Coalition** to access climate finance and technology for moving away from fossil fuels.

Conclusion:

India's heatwave crisis reflects a **deepening climate vulnerability** with serious implications for **health, livelihoods, and food security**. Current responses, largely **reactive and underfunded**, fail to address structural causes. Without **long-term investment**, **proactive governance**, and **global cooperation**, parts of India may approach **human survivability limits**.

Mains Practice

Q. The increasing frequency and intensity of pre-monsoon heatwaves in India are not merely a meteorological phenomenon but a complex socio-economic crisis. Analyze the structural limitations of India's Heat Action Plans (HAPs) and suggest comprehensive measures to mitigate this crisis. (250 Words, 15 Marks)

Scaling Climate Adaptation From Policy to Grassroots

GS III: Environmental Pollution and degradation

Context:

India is the **9th most climate-vulnerable country** in the world. Between 1995 and 2024, the country faced 430 extreme weather events, causing **$170 billion in economic losses** and affecting 1.3 billion people.

Key Concepts:

- **Mitigation:** Reducing the root causes of climate change (e.g., solar energy, electric vehicles).
- **Adaptation:** Accepting that climate change is happening and preparing to handle its consequences (e.g., drought-resistant seeds, building dikes).

What are NDCs?

- **Nationally Determined Contributions** — each country's climate commitments under the Paris Agreement

India's Updated NDCs (2031–35):

- **Climate resilience** to be mainstreamed into every policy
  - *Mainstreaming: every development project must include a climate angle*
- Focus on **coastal resilience, disaster preparedness & heat mitigation**
- **Biodiversity** and **sustainable livelihoods** integrated

Global Targets:

- **Triple adaptation finance by 2035**
- **COP30 → Adopt Belém Adaptation Indicators** to measure

Climate Finance Crisis & Taxonomy Gap

The Finance Problem:

- **FY22:** India spent only **5.6% of GDP on adaptation**
- **Union Budget 2026–27 is mitigation-skewed** (EVs, solar)
- **UNEP Adaptation Gap Report 2025: $284–339 Billion annual gap** for developing countries

Climate Finance Taxonomy — What is it?

- A classification system defining which projects are **'green'**
- **India's Draft Climate Finance Taxonomy (2025)** released

Critical Gap in India's Taxonomy:

- Focuses only on **mitigation** (emission avoidance)
- **Adaptation projects are NOT covered in the Taxonomy**

Key Government Initiatives

- **NICRA (National Innovation in Climate Resilient Agriculture):**
  - Launched by the **Indian Council of Agricultural Research (ICAR)**.

- Identified 151 vulnerable hotspots across 651 districts.
- **Key Actions:** Providing drought-resistant seeds, promoting drip irrigation, encouraging crop diversification, and farmer capacity building.

- **Climate Resilient Villages (CRV) - The Tamil Nadu Model:**
  - Highlighted in the **Economic Survey 2026**, this model covers 11 vulnerable districts with support from the World Resources Institute (WRI).
  - **Four Pillars:** Local water management, renewable energy promotion, biodiversity conservation, and livelihood security.
  - **Bottom-Up Approach:** Local communities are directly involved in the decision-making process.
- **Updated NDCs:** India aims to triple its climate adaptation targets by 2035, focusing on coastal resilience and disaster preparedness.

Challenges

- **Financial Gap:** In 2022, India spent only **5.6% of its GDP on adaptation**. Globally, developing countries face a massive adaptation funding gap of $284 to $339 billion.
- **Mitigation Bias:** Most government and private funding currently targets mitigation projects (like green energy) because they are easier to categorize as "green" investments, leaving adaptation underfunded.
- **Centralized Planning:** Top-down policies often fail to address specific local challenges.
- **Global Shortfall:** UNEP estimates a funding gap of **$284 to $339 billion** for adaptation in developing countries.
- **Institutional silos:** National and State adaptation plans often remain centralized and lack integration with local grassroots initiatives.

Way Forward

- **Locally Led Adaptation (LLA):** Empowering **Panchayati Raj institutions** and Urban Local Bodies to lead climate planning, as they understand local vulnerabilities best.
- **Institutional Changes:** Creating **Climate Change Cells** at every department and district level.
- **Financial Reforms:** Attracting private sector investment and ensuring budget reallocations prioritize adaptation.
- **Integrated Planning:** Combining centralized policy (Top-Down) with decentralized community participation (Bottom-Up) to create a comprehensive resilience framework.

Mains Practice

Q. India's updated NDCs emphasize mainstreaming climate adaptation, yet the financial and institutional frameworks remain mitigation-centric." Analyze the statement and suggest measures for scaling Locally Led Adaptation (LLA). (250 Words, 15 Marks)

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